

4. INDUSTRIAL REVOLUTION

- Industrial Revolution, in modern history, the process of change from an agrarian, handicraft economy to one dominated by industry and machine manufacture.
- This process began in England in the 18th century and from there spread to other parts of the world.
- Although used earlier by French writers, the term Industrial Revolution was first popularized by the English economic historian Arnold Toynbee (1852–83) to describe England's economic development from 1760 to 1840. Since, Toynbee's time the term has been more broadly applied.
- If French Revolution is one of the two milestones of the modern world history, the second one is Industrial Revolution.
- While French Revolution ushered decisive changes in socio-political fields, Industrial Revolution opened a new era by bringing fundamental changes in economy and production.
- Both the revolutions did not confine themselves to the fields and countries they have occurred but spread to other areas and parts of the world like the waves spreading all corners of the pond.
- Both the revolutions have changed the nature and course of the human history.
- Industrial revolution was the major radical change that took place in the field of production and the revolution was made possible through science and technology.
- Industrial revolution means machine taking the place of human crafts, products of machines substituting the hand made products. Power generating sources like water, coal and electricity taking the place of human labour and skill was the industrial Revolution.
- Industrial Revolution shook the very foundations of the Production that continued unchanged for nearly two thousand years.
- Power loom replacing hand loom, steel replacing wood, Trains replacing horse rides, power produced by burning charcoal replacing the energy produced by the workingmen was the industrial Revolution.
- Industrial Revolution that brought the revolutionary change in the mode of production has begun in second half of 18th century and still continuing.
- Hence, historians tend to call it industrial revolution. However, since time is relative and technological development cannot be achieved in a day or two, achieving the same in one or two centuries can

- Science and society going hand in hand in western world made possible the industrial Revolution to take place in the west.
- The philosopher, the scientist and the artisan lived together without any social discrimination in western society.
- Even though the origins of science lie in eastern world, the social equality persisting in western world was the reason behind the development in science and civilization there.
- During ancient and medieval ages the artisan or the 'worker paid his attention in creating a product, but never bothered to understand the scientific reason or theory behind it.
- The artisan satisfied himself with the technical knowledge, but could not perceive the scientific knowledge.
- However, the enlightenment of the Renaissance period had wiped out the demarcating line between the worker, who does the physical work and the scientist who does the mental exercise.
- During renaissance period the handicrafts have fallen into the hands of those who were liberated and enjoyed status and prestige in the society.
- They become closer to the ruling sections.
- It was believed that the physical labour was associated with slavery and it was nothing to do with creativity.
- However the Renaissance has shattered the dogma and allowed the two to go farther resulting in the invention of wonderful devices which made Industrial Revolution possible.

Factors leading to the beginning of Industrial Revolution in England

- Industrial Revolution made its beginning in England during 18th century. Certain factors existing in England were favourable to it.
- **The first being capital.** The capital required for the industrialization was abundant in England.
- The capital accrued as a result of agricultural and commercial revolutions that happened. 17th, 18th centuries in England facilitated the establishment factories, purchase of raw material and appointment of workers.
- Bank of England, London stock exchange and British currency helped the consolidation of British economy. Adding to its England had enough enthusiastic and courageous merchants and feudal lords, entrepreneurs who dared to make investments in new ventures.
- **The second factor was labour.** The needed workers to work in the certainly be considered as achieved in a very short period.
- Hence it is more appropriate to call it industrial Revolution'. Louis Blanc- the French socialist was the first to call the developments taking place in the

field of machines and production as Industrial Revolution.

- However Arnald Toynbee, the British historian made the phrase popular by writing a book with the title 'Industrial Revolution'.the factories were in England, as the population grew rapidly in 18 century and the same was doubled in 19th century England.
- Even though many of them migrated to British got incorporated into factories, Peasants who lost their fields due to the enclosure movement also reached towns and cities and settled as factory workers, the disturbed political conditions in various European countries also helped the common people to reach England and made themselves available for cheaper wages.
- **The third factor was resources.** England was well endowed with resource. Its climate was damp enough be suitable for mechanical spinning and weaving in Cool climate was always an asset for spinning as the thread will not easily get snapped. Water resources were ample. Iron and coal mines were abundant.
- **The fourth factor was transportation.** By the beginning of 18th century England was leading the world in shipping and seaborne transport. England modernised its medieval internal roads and canals. England being an island nation with suitable geographical condition to form natural ports, had the coastal line not more than 60, 70 miles distant from town or city inside the country, and helped it grow in internal and external trade.
- **The fifth factor was markets.** As the British colonies have extended to Asia and Africa, extensive markets were opened to it. At this point a special mention should be made to Indian market. Adding to its internal open market among England, Scotland and Ireland has been consolidated. Hence there was no scarcity in England either for the supply of raw or for the markets for finished products.
- The above factors helped the industrial revolution to begin and spread in England. The following were some of the important industries, the growth of which helped the outbreak of industrial revolution.

Text industry

- Industrial revolution had its beginning in textile industry. Spinning and weaving did exist since the ancient civilization.
- However, weaving continued as a handicraft. Spinning was very slow. One weaver used to exhaust the yarn spun by five persons.

- The first revolutionary step took place when John Kay invented the flying shuttle in 1733. Flying shuttle doubled the speed of weaving which resulted in the shortage of yarn.
- Hence, spinning jenny was invented by James Hargreaves in 1764 that can draw thin and fine thread from 16 spindles.
- However, the thread drawn from the Jenny was not strong enough which resulted in the invention of water frame by Richard Arkwright in 1769. Water Frame consisted of series of rollers and was run by power. Yarn spun on water frame was hard enough 'not 80 fine.
- Hence combining the advantages of spinning jenny and water frame, Samuel Crompton invented the spinning mule in 1779. The, Mule facilitated spinning hundreds of yarn at a time.
- The spinning mule also didn't work properly under new methods. Hence Edmund Cartwright invented the power loom.
- Through which mechanism was applied to weaving which resulted in speedy production. Spinning mule and power loom brought revolutionary changes in textile industry of England.

Steam Engine

- The real revolution took place when steam was introduced as power in place of human energy and Waterpower.
- It is largely believed that James Watt had invented the steam engine.
- However, Newcomen was the precursor of Steam engine.
- But it was restricted to pumping out the water from coal mines. As it could not pump out water from deeper mines, James Watt rectified it and invented steam engine in 1799.
- This steam engine has widely come into use. It worked as the driving force for the textile industry, trains and steamers.
- Bolton has widely propagated the use of steam engine and succeeded in producing them in large numbers. He had even invented new machines using steam engine and succeeded in producing them in large numbers.
- He had even invented new machines using steam engine. These developments led to the closure of domestic production and establishment of heavy machinery and industries.

Iron and Steel industry

- A huge amount of iron and steel was required for manufacturing machines. To process iron, charcoal was needed.
- Earlier to the Industrial Revolution iron was processed in small furnaces with the help of charcoal.
- To acquire charcoal, woods were burnt. Hence rock coal replaced charcoal. Henry Cort had invented the puddling process with the help of rock coal. As a result of it rock coal was much needed.
- It has become dangerous to dig out rock coal from the mines because of poisonous gases existing in mines.
- It has become impossible to take, lamps inside the mines because of the prevalence of combustible gases. Under these circumstances Humphry Davy invented safety lamp and with it were mines illuminated and accidents were avoided.
- As the production of coal increased, the iron and steel industry was also developed. The greatest contribution to the development of steel industry was made by Bessemer.
- He had invented a process from steel which called Bessemer method. Bessemer had established a big steel factory in England. Darby, Hutsman, Nicholson were others who contributed to steel industry.
- Steel was extensively used in manufacturing of machines, railway lines; steamers and various instruments.

The Impact of Industrial Revolution

- The industrial Revolution had shown immense impact on the human history. The results of the Revolution have spread to economic, political, social, scientific and technological fields bringing fundamental change.

Economic impact

- The Industrial Revolution has brought many a change in the field of economy since it is the very outcome of it.
- Industrial Revolution replaced the domestic system with factory system. Machines have been established in factories.
- Factories needed well constructed spacious establishments.
- Money needed for establishing factories, purchasing raw materials and payments of wages to workers.
- Hence an industrial capitalist system has evolved that can invest money as capital.
- Consequently the feudal lords and the entrepreneurs of domestic system who

saved money and the merchants benefited by mercantilism have turned to be capitalists.

- Corporations and joint stock companies have emerged as it was not possible to invest capital by one or two for the big factories.
- Hence establishment of huge factories has gone into the hands of joint companies.
- Capitalists were kept away from the problems of trade as the managerial staff had looked after them.
- European countries had to acquire markets for the enormous production made as a revolution.
- This led to the establishment of colonies all over the world. The colonies have turned to be the markets to the finished products manufactured by the raw materials supplied by them.
- Colonisation resulted in disputes Among European countries finally leading to World War I. While industrial revolution led to colonialism, colonialism resulted in imperialism.
- Industrial revolution had created new goods and articles which did not exist in medieval ages.
- Means of transport and means of communication like telegraph and telephone have made human life more convenient and comfortable.
- Yet another result of the industrial Revolution was the rise of economic liberalism. Free trade policy introduced in England, based on the policy of “Laissaz faire” was the basis of the economic liberalism.
- According to it the Government shall not interfere in the process of production and distribution. Economic liberalism has championed the cause of individual ability in economic progress.
- This has attracted the British capitalists very much. However the British capitalists in their attempt to stop the government intervention in the trade policy didn't allow the factory acts intended for workers welfare to be promulgated.
- Industrial Revolution has spread to other European countries after 1870. With the completion of unification of Italy and Germany, the nationalistic sentiment has spread to other parts of the globe.
- Under the circumstances to compete with England, merchants of certain countries needed the intervention of government in the trade policy.
- They needed protection of taxes.
- Hence, the economic liberalism was replaced by economic nationalism through which national interests have been in trade policies.

Social impact

- Industrial Revolution caused greater social transformation to take place.

- Towns that flourished in ancient and medieval ages have lost glory, giving place to new cities developed in and around factories and industries, conveniently located for the supply of raw material, power generators and means of transportation.
- Industrial Revolution witnessed the rise of two classes in these new cities.
- They are 1) capitalist class and 2) Working class. Peasants who have lost their fields as a result of enclosure movement have reached the cities that turned as workers.
- Fascination towards industries also drove some people to the cities, which joined the working class.
- The people from different areas flocking together led to the rise heterogeneous culture.
- The customs, tradition, moral values and the very life style of the people underwent a great change.
- As the socio-cultural remnants of ancient and medieval ages are disappearing the dazzling picture of modern life was unveiling.
- The products of the cottage industry could not compete with the machine products as the machine products were finer and cheaper. Consequently the cottage industry has collapsed.
- As result of it, skilled workers and artisans in the rural areas had to move to cities to work in factories.
- The artisans who hitherto enjoyed individual status and respect had to surrender to the capitalist.
- As the workers settled in huge numbers in cities their living places have turned as slums, where hunger and unhygiene prevailed. This also gave rise to problems of housing, water scarcity etc.
- The life worker has become wretched. Adding fuel to fire, the increase in the number of workers has decreased their wages.
- Workers had to work under conditions not congenial and hygienic. They had to work in places without proper passage of air and sufficient light. They didn't have fixed duration of work.
- Women and children were also forced to drudgery. There was no one who cared for the education and health of the worker and his family.
- An exception to the above said situation was Robert Owen of Scotland who

laid foundations to the naturalism with his welfare activities to workers.

- Owen has reduced the working hours of the workers, stopped engaging children in work. He had created hygienic conditions in the work place and even shared the profits with the workers.

Political impact

- Industrial revolution had its impact on political system and world politics.
- As a result of the results revolution new systems like joint stock companies, workers association, rail companies Municipal Corporation have evolved.
- The legislature and the administration had to deal with the new institutions. Monarchical form of government that were working under the purview of ancient systems and traditions, failed to solve the problems of the new institutions.
- As the newly emerged classes realised that the existing political system cannot meet the new requirement and demands, revolts and movements took place.
- Government with strong administrative system and public support alone could solve the problems posed by new institutions.
- Industrial revolution resulted in the emergence of a new class in society came to be known as middle class. This class consisted of the merchants and the capitalists.
- Emerged with self potential, this class was different from the medieval feudal section.
- Being affluent, this section demanded their share of political representation.
- They wanted the age old institution and laws be abolished.
- They even demanded their rightful social recognition as factory owners who provided jobs to millions of people.
- As a result of it monarchical administration had to undergo certain changes. The reforms act of 1832 has transformed certain political powers to the middle class from the ecclesiastical and feudal sections in England.
- Working class, devoid of education and medical facilities, realised that their demands shall be achieved only through pressurising the political sections.
- Hence they demanded right to vote and right to form associations. As the working class and middle class demanded their political rights, fearing the working class, the ruling class, allowed the middle class to acquire their share in the government.

- As a result of the above developments revolutionary changes took place in the aims and objectives of the governments. Government had to take care of the socio economic status of the people.
- New welfare states had to emerge, doing away with the dogma of the ruler and the ruled; giving way to the concept of the government and the citizens and it is proved that the government and the society are interdependent.

Intellectual impact

- Industrial revolution paved way for the rise of new ideas, concepts and economic theories that influenced a great deal the history of modern world forward by socialism.
- Socialism has condemned the unequal distribution of wealth in the society. It had declared that wealth is the product of combined effort of capital and labour.
- Then uneven distribution of wealth is the main cause for the economic inequalities.
- France has become the birth place of socialism. Proudhon and Louis Blanc were the earliest socialists of France.
- Karl Marx has modernised the socialist theories and propounded the communism. Karl Marx, a German together with Frederick Engels of England had published what is known as “communist manifesto” in 1848.
- It was a foundation of world’s political and economic systems.
- Karl Marx has given a new interpretation to history by saying that “History of all hitherto existing society is history of class struggles” the classes being the haves and have-nots.
- While the clash in ancient period was between the chief and slave, in the medieval age between feudal lord and peasant, in modern times it will be between the capitalist and workers that leads to the establishment of society which is classless.
- The influence of communist theory of Karl Marx had spread to all over Europe.
- Consequently countries like Germany and France have introduced welfare measures to workers.
- Governments have taken care of education, health, hygiene, insurance and pensions of the workers. England promulgated certain legislation supporting the working class.
- The emergence of the two opposite political and economic systems namely the

capitalism and the communism is the paradoxical and phenomenal result of the industrial revolution.

- The industrial revolution that started in England brought the fundamental change in production, spread to Europe initially, later to entire world, influenced all walks of life, making it comfortable to some, wretched to many others, led to rise of capitalism on one side and exploitation on the other side.
- Capitalism leading to imperialism, imperialism to world wars and exploitation to socialist revolution.
- Intellectual of the world have proposed new economic theories that bring change to the wretched conditions of the workers.
- The Humanists, utopians socialists and communists were among them.
- The theories of socialists and communists have received greater attention. As social equality was proposed by democracy, economic equality was brought